

Elle Miller

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in [ellemiller101](#) 🏠 [google scholar](#)

EDUCATION

2023 – present

Ph.D. in Computer Science

UNIVERSITY OF EDINBURGH

Thesis: “*Tactile representation learning in reinforcement learning for robot control*”.

2017 – 2022

B.Eng. in Mechatronics (Space) &

B.Sc. in Advanced Physics

UNIVERSITY OF SYDNEY

Grade: High Distinction

Thesis: “*Learning with less labels for off-road semantic segmentation*”, undertaken at NASA JPL.

PUBLICATIONS

First author

1. *Enhancing Tactile-based Reinforcement Learning for Robotic Control*
EM, Trevor McInroe, David Abel, Oisín Mac Aodha, and Sethu Vijayakumar. NeurIPS 2025.
 - Spotlight presentation: Dexterous Humanoid Manipulation Workshop at Humanoids 2025.
2. *Unknown Object Grasping for Assistive Robotics*
EM, Maximilian Durner, Matthias Humt, Gabriel Quere, Wout Boerdijk, Ashok M. Sundaram, Freek Stulp, and Jorn Vogel. ICRA 2024.
3. *The Formation of Wide exoKuiper Belts from Migrating Dust Traps*
EM, Sebastian Marino, Sebastian Stammer, Paola Pinilla, Christian Lenz, Til Birnstiel, and Thomas Henning. MNRAS 2021.

Co-author

1. *Learning Long-Horizon Robot Manipulation Skills via Privileged Action*
Xiaofeng Mao, Yucheng Xu, Zhaole Sun, **EM**, Daniel Layeghi, Michael Mistry. CoRL 2025.
2. *Few-shot Semantic Learning for Robust Multi-Biome 3D Semantic Mapping in Off-Road Environments*
Deegan Atha, Xianmei Lei, Shehryar Khattak, Anna Sabel, **EM**, Aurelio Noca, Grace Lim, Jeffrey Edlund, Curtis Padgett, Patrick Spieler. ACRA 2024.

EXPERIENCE

2025 summer

Visiting PhD Researcher

WASEDA UNIVERSITY

Investigated and implemented sim-to-real transfer of reinforcement learning policies for caregiving tasks on the AIREC humanoid robot.

2023

Robotics Research Engineer

DLR INSTITUTE FOR ROBOTICS AND MECHATRONICS

Designed an assistive manipulation system to promote user agency in automated grasping of unknown objects with the EDAN robot, resulting in an ICRA publication.

2022

Visiting Student Researcher

NASA JET PROPULSION LABORATORY

Developed methodologies to achieve reliable visual segmentation of off-road terrains from small labelled datasets as part of the DARPA RACER Challenge. I was selected by NASA to join the JVS RP programme through a University of Sydney scholarship.

2022	Robotics Research Intern	UNIVERSITY OF EDINBURGH
	Integrated a dual-armed torso robot with an omnidirectional mobile base in software (e.g. sensor drivers, motor controllers) and implemented whole-body control for human-robot interaction.	
2020 – 2021	Astrophysics Research Intern	MAX PLANCK INSTITUTE FOR ASTRONOMY / UNI OF CAMBRIDGE
	Proposed a new theory of debris disc formation through the study of planetesimal formation in simulation. I was 1 of 5 interns to be selected internationally, and I continued the research as my supervisor moved institutions and I was a full-time student. The work entailed 1000+ simulations which resulted in a first-author paper.	
2020 summer	Firmware Engineering Intern	COCHLEAR
2019 summer	Research Scholar in Cosmology	AUSTRALIAN NATIONAL UNIVERSITY
2019	Space Engineering Intern	SABER ASTRONAUTICS

HONORS

2025	Invitation, Tactile Sensing and Processing Summer School 2025	DRESDEN
2025	Longlisted for Student Tutor of the Year	UNIVERSITY OF EDINBURGH
2025	Best talk at the annual CDT conference	EDINBURGH CENTRE FOR ROBOTICS
2024	Best talk at the annual CDT conference	EDINBURGH CENTRE FOR ROBOTICS
2022	Engineering Student Industry Placement Scholarship	NASA JPL & UNIVERSITY OF SYDNEY
2022	Dean's Excellence in Academic Performance	UNIVERSITY OF SYDNEY
2017 – 2022	Sydney Scholars Award	UNIVERSITY OF SYDNEY
2016	Proxime (ATAR 99.6, top 0.6% of state)	SECONDARY SCHOOL

SKILLS

Technical: Reinforcement learning, deep learning, robotics.

Tools: Languages (Python, C++, Matlab, URDF), middleware (ROS), simulators (Isaac Lab, MuJoCo, Gazebo), misc (PyTorch, Slurm, WandB).

Soft: Critical thinking, writing, public speaking, leadership, positivity, resilience

TEACHING

Teaching Assistant	Robot and Reinforcement Learning INFR11285	UNIVERSITY OF EDINBURGH
Teaching Assistant	Reinforcement Learning INFR11010	UNIVERSITY OF EDINBURGH

SERVICE

2024 – present	Postgraduate Student Representative	INSTITUTE FOR PERCEPTION, ACTION, AND BEHAVIOUR, UoE
2019	Social Secretary	EDINBURGH UNIVERSITY MECHANICAL ENGINEERING SOCIETY
2019	Events Director	SYDNEY UNIVERSITY MECHATRONICS ORGANISATION
2018	Secretary	SYDNEY UNIVERSITY PHYSICS SOCIETY
2017 – 2018	1st & 2nd Year Representative	SYDNEY UNIVERSITY ROCKETRY TEAM